

Pressure ulcer

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Definition

Pressure ulcer is “an injury to the skin and/or underlying tissue resulting from pressure or in combination with shear, usually over a bony prominence”.

- As tissue is compressed, blood is diverted and blood vessels are forcibly constricted by the persistent pressure on the skin and underlying structures; thus cellular respiration is impaired and cells die from ischemia and anoxia.
- A pressure ulcer is localized damage to the skin and underlying tissue due to unrelieved pressure, often compounded by shear forces, typically over bony prominences. Persistent pressure disrupts blood flow, causing ischemia, anoxia, and eventual cell death.
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Prevalence

Older people account for 70% of all pressure ulcers. Pressure ulcers occur in all settings across the continuum, with the highest incidence reported in hospitalized vulnerable older people undergoing orthopedic procedures (9% to 19%) and individuals with quadriplegia (33% to 60%).

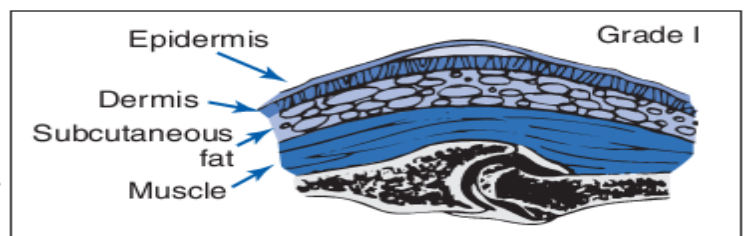
Classification

Pressure ulcer development

Stage I : REVERSIBLE WITH INTERVENTION.

Erythema not resolving within thirty (30) minutes of pressure relief.

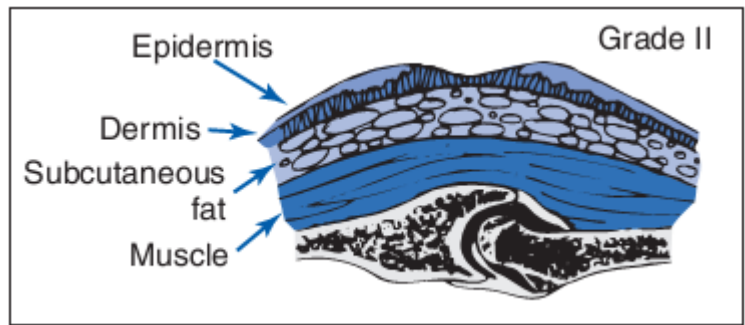
Epidermis remains intact.



Stage II

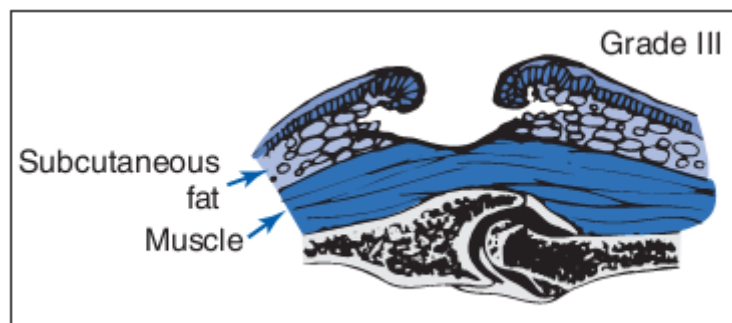
Partial-thickness loss of skin layers involving epidermis and possibly penetrating into but not through dermis.

May present as blistering with erythema and/or induration; wound base moist and pink; painful; free of necrotic tissue.



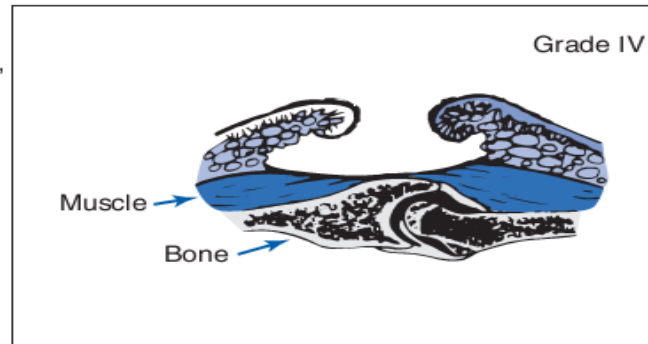
Stage III

Full-thickness tissue loss extending through dermis to involve subcutaneous tissue. Presents as shallow crater unless covered by eschar. May include necrotic tissue, undermining, sinus tract formation, exudate, and/or infection. Wound base is usually not painful.



Stage IV

Deep tissue destruction extending through subcutaneous tissue to fascia, possibly involving muscle layers, joint, and/or bone. Presents as a deep crater. May include necrotic tissue, undermining, sinus tract formation, exudate, and/or infection. Wound base is usually not painful.



Classification

1. **Stage I:** Reversible erythema; intact epidermis.
2. **Stage II:** Partial-thickness loss; possible blistering; no necrotic tissue.
3. **Stage III:** Full-thickness loss; may involve necrosis and infection.
4. **Stage IV:** Deep tissue damage; potential bone/muscle involvement.

Risk Factors

- Many factors increase the risk of pressure ulcers in older adults including changes in the skin, comorbid illnesses, nutritional status, cognitive deficits, and reduced mobility.
- “The primary risk factors for pressure ulcer development are immobility and limited activity with positioning that exerts unrelieved pressure on tissue confined between nonpliable surfaces”.
- Tissue tolerance, in addition to unrelieved pressure, contributes to the risk of a pressure ulcer. Tissue tolerance is related to the ability of the tissue to distribute and compensate for pressure exerted over bony prominences. Factors that affect tissue tolerance include moisture, friction, shear force, nutritional status, age, sensory perception, and arterial pressure.

Risk Factors

- **Primary Risks:** Immobility, unrelieved pressure.
- **Contributing Factors:** Tissue tolerance affected by:

- **Moisture:** Macerates skin.
- **Friction and Shear:** Damage from movement.
- **Nutritional Deficiencies:** Impaired healing.
- **Age:** Reduced skin resilience.
- **Comorbidities:** Decreased systemic resilience.

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Consequences

- Pressure ulcers are costly to treat and prolong recovery and extend rehabilitation.
- Pressure ulcers had a profound impact on the patients' lives, physically, socially, emotionally, and mentally.
- Complications include the need for grafting or amputation, sepsis, or even death, and may lead to legal action by the individual or his or her representative against the caregiver.

Implications for Gerontological Nursing and Healthy Aging

Assessment Key Aspects of Assessment of a Pressure Ulcer

- 1. Measure and document wound dimensions.**
- 2. Assess surrounding tissue, edges, and wound bed.**
- 3. Evaluate exudate characteristics.**

1. Location and exact size (width, depth, length)
2. Condition of the surrounding tissue
3. Condition of the wound edges: for example, smooth and white or irregular and pink
4. Wound bed: warmth, moisture, color, odor, amount, and color of exudate

Interventions

1. The goal of nurses is to help maintain skin integrity against the various environmental, mechanical, and chemical assaults that are potential causes of breakdown.
2. In promoting healthy aging of all persons, nurses focus on prevention, taking action to eliminate friction and irritation to the skin, such as from shearing; to reduce moisture so that tissues do not macerate
3. Displace body weight from prominent areas to facilitate circulation to the skin.
4. The nurse should be familiar with the types of supportive surfaces and types of dressings so that the most effective products are used.
5. The nurse should assess the frequency of position change, adding pillows so that skin surfaces do not touch, and establish a turning schedule if needed.
6. Nutritional intake should be monitored, as well as the serum albumin, hematocrit, and hemoglobin levels. Caloric, protein, vitamin, and/or mineral supplementation can be considered if there is evidence of deficiencies of these nutrients.
7. Full discussion of the treatment options and indications for their use in the nursing care and treatment of pressure.
8. The type of dressing selected is based on the condition of the ulcer; the presence of granulation, necrotic tissue, and slough; the amount of drainage; microbial status; and the quality of the surrounding skin.
9. “Too frequently, aggressive and expensive interventions are implemented without attention to basic care practices such as provision of adequate nutrition, good hygiene, and proper positioning”.
10. Provision of education to patients, families, and professional staff must also be included in any skin care program.

11. Consultation with a wound care specialist is advisable for wounds that are extensive or non-healing. Specialized nurses such as enterostomal therapists or nurse practitioners may work with wound centers or surgeons to provide consultation in nursing homes, offices, or clinics.

12. Promoting Wound Healing

- Keep wound warm at all times.
- Keep clean and moist at all times.
- Protect from further injury.
- Promptly absorb exudate and fill dead space with a biofriendly material.
- Do not subject tissue to caustic products (e.g., Betadine)

1. Preventive Care:

- Reposition patients regularly.
- Use pressure-relieving devices.
- Minimize moisture and friction.
- Monitor nutritional intake (protein, vitamins, minerals).
- Maintain serum albumin and hemoglobin levels.

2. Skin and Wound Care:

- Clean and keep the wound moist.
- Avoid caustic agents like Betadine.
- Use appropriate dressings based on ulcer characteristics.

3. Patient and Staff Education:

- Teach prevention strategies.
- Provide detailed treatment plans.

4. Collaborative Care:

- Consult wound care specialists for complex cases.

Promoting Healing

- Maintain warmth and moisture at the wound site.
- Protect against further injury.
- Absorb exudate effectively.
- Fill dead spaces with biofriendly materials.